Antonio Alcántara Mata

Tenure-track Assistant Professor

Department of Quantitative Methods, CUNEF Universidad
Campus Almansa, Building D, Room 4.29 · Calle Almansa 101, 28040 Madrid, Spain

antonio.alcantara@cunef.edu · (+34) 665 019 751 · antonioalcantara.com · ORCID 0000-0002-3306-2246

ACADEMIC APPOINTMENTS

Sep. 2026 – Present — **CUNEF Universidad** — *Spain*
Tenure-track Assistant Professor, Quantitative Methods Department
Researching on machine learning and data-driven optimization for decision-making under uncertainty, with a particular interest in energy/power systems, alongside teaching courses in analytics and operations research.

Sep. 2025 – Aug. 2026 — **Technical University of Denmark (DTU)** — *Denmark*
Postdoctoral Researcher, Department of Wind and Energy Systems
Advisor: Prof. Spyros Chatzivasileiadis
Foundation models for power systems, with a focus on uncertainty quantification and trustworthiness.

Dec. 2021 – Jun. 2025 — **Universidad Carlos III de Madrid (UC3M)** — *Spain*
Predoctoral Researcher (FPU Fellow), Department of Statistics
Advisor: Prof. Carlos Ruiz Mora
Machine learning and operations research, focusing on the use of ML models as surrogates in optimization problems, such as neural network MILP reformulations.

Jan. 2021 – Dec. 2021 — **Universidad Carlos III de Madrid (UC3M)** — *Spain*
Research Assistant, Department of Computer Science
Advisor: Prof. Inés M. Galván
Probabilistic forecasting models for regional renewable energy generation.

RESEARCH STAYS

Aug. 2024 – Oct. 2024 — **KTH Royal Institute of Technology** — *Sweden*
Visiting Researcher, Department of Mathematics
Host: Prof. Jan Kronqvist

Sep. 2023 – Dec. 2023 — **Imperial College London** — *United Kingdom*
Visiting Researcher, Department of Computing
Host: Prof. Calvin Tsay

EDUCATION

May 2025 — **Universidad Carlos III de Madrid (UC3M)** — *Spain*
Ph.D., Mathematical Engineering
Advisor: Prof. Carlos Ruiz Mora
Dissertation: “Machine Learning for Decision-making under Uncertainty: New Methodologies and Energy Applications”

Jul. 2021 — **Universidad de Granada (UGR)** — *Spain*
M.S., Applied Statistics

Jul. 2020 — **Universidad Carlos III de Madrid (UC3M)** — *Spain*
B.S., Statistics and Business

RESEARCH INTERESTS

- Constraint learning and surrogate models: embedding machine learning models into optimization problems
- Uncertainty quantification: prediction intervals, quantile regression and conformal prediction
- Trustworthy AI/ML for power and energy systems

PUBLICATIONS

Journal Articles (10)

[1] **Risk-averse wind farms placement via quantile constraint learning.**
Feng, W., **Alcántara, A.**, & Ruiz, C. (2026). *Computers & Operations Research.* doi:10.1016/j.cor.2026.107584

[2] **Integrated design and scheduling of hydrogen processes under uncertainty: a quantile neural network approach.**
Ghilardi, L. M., Patrón, G. D., **Alcántara, A.**, & Tsay, C. (2025). *Industrial & Engineering Chemistry Research.* doi:10.1021/acs.iecr.5c03288

[3] **Leveraging neural networks to optimize heliostat field aiming strategies in concentrating solar power tower plants.**
Alcántara, A., Diaz-Cachinero, P., Sánchez-González, A., & Ruiz, C. (2025). *Energy and AI.* doi:10.1016/j.egyai.2025.100520

[4] **A quantile neural network framework for two-stage stochastic optimization.**
Alcántara, A., Ruiz, C., & Tsay, C. (2025). *Expert Systems with Applications.* doi:10.1016/j.eswa.2025.127876

[5] **Optimal day-ahead offering strategy for large producers based on market price response learning.**
Alcántara, A., & Ruiz, C. (2024). *European Journal of Operational Research.* doi:10.1016/j.ejor.2024.06.038

[6] **A neural network-based distributional constraint learning methodology for mixed-integer stochastic optimization.**
Alcántara, A., & Ruiz, C. (2023). *Expert Systems with Applications.* doi:10.1016/j.eswa.2023.120895

[7] **On data-driven chance constraint learning for mixed-integer optimization problems.**
Alcántara, A., & Ruiz, C. (2023). *Applied Mathematical Modelling.* doi:10.1016/j.apm.2023.04.032

[8] **Deep neural networks for the quantile estimation of regional renewable energy production.**
Alcántara, A., Galván, I. M., & Aler, R. (2023). *Applied Intelligence.* doi:10.1007/s10489-022-03958-7

[9] **Pareto optimal prediction intervals with hypernetworks.**
Alcántara, A., Galván, I. M., & Aler, R. (2023). *Applied Soft Computing.* doi:10.1016/j.asoc.2022.109930

[10] **Direct estimation of prediction intervals for solar and wind regional energy forecasting with deep neural networks.**
Alcántara, A., Galván, I. M., & Aler, R. (2022). *Engineering Applications of Artificial Intelligence.* doi:10.1016/j.engappai.2022.105128

Working Papers (1)

[11] **Trustworthiness layer for foundation models in power systems: application to N-k contingency screening.**
Alcántara, A., & Chatzivasileiadis, S. (2026). *arXiv:2602.07995*

TEACHING EXPERIENCE

Fall 2026	**Data Analysis** B.S. in International Business and Economics (1 group) Main lecturer	*CUNEF*
2022–2025	**Statistics for Engineering** B.S. in Mechanical, Electrical, Industrial and Aerospace Engineering (4 groups) Co-course responsible for lectures, exam design and grading Student evaluations: 4.38, 4.52, 4.75 and 4.92 (out of 5)	*UC3M*
2023–2024	**Introduction to Statistical Modeling** B.S. in Data Science and Communication Engineering (2 groups) Co-course responsible for lectures, exam design and grading Student evaluations: 4.92 and 5.00 (out of 5)	*UC3M*

GRANTS & AWARDS

2021–2025	FPU Fellowship, Spanish Ministry of Science and Universities *Highly competitive national fellowship (15 awarded nationally in field)*
2022–2024	Teaching Excellence Acknowledgement, UC3M
2023	Competitive Mobility Grant, UC3M
2020	Extraordinary End of Studies Award, UC3M
2019	Social Council Excellence Award, UC3M

2019	Santander Progreso Scholarship, Santander Bank
2018	Excellence Award, Regional Government of Madrid

SELECTED TALKS

Invited Seminars

Oct. 2026	6th Conference on Foundation Models of the Electric Grid (GridFM), Trusted Grid AI session, Porto, Portugal *A Trustworthiness Layer for GridFM via Conformal Prediction Intervals: Implementation and Open Questions.*
Apr. 2026	GridFM Working Group *Trustworthiness Layer for Foundation Models in Power Systems.*
Mar. 2026	International System Operator Network (ISON) *Trustworthiness Layer for Foundation Models in Power Systems.*
Feb. 2026	CUNEF Universidad, Quantitative Methods Department, Spain *Embedding Machine Learning into Optimization: Strategic Decision-making in Energy Markets.*
Oct. 2024	Linköping University, Dept. of Mathematics, Sweden *A Quantile Neural Network Framework for Two-stage Stochastic Optimization.*
Sep. 2024	KTH Royal Institute of Technology, Dept. of Mathematics, Sweden *A Quantile Neural Network Framework for Two-stage Stochastic Optimization.*
Sep. 2023	Imperial College London, Dept. of Computing, United Kingdom *Probabilistic Forecasting, Quantile Neural Networks, and Stochastic Optimization.*

Conference Presentations

Jul. 2026	24th Conference of the International Federation of Operational Research Societies (IFORS), Vienna, Austria *Leveraging neural networks to optimize heliostat field aiming strategies in Concentrating Solar Power Tower plants.*
Jun. 2025	34th European Conference on Operational Research (EURO), Leeds, United Kingdom *From Sample Average Approximation to Surrogate Models in Smart Grids Operation.*
Jun. 2025	41st National Congress of Statistics and OR (SEIO), Lleida, Spain *A Quantile Neural Network Framework for Two-stage Stochastic Optimization.*
Jul. 2024	33rd European Conference on Operational Research (EURO), Copenhagen, Denmark *A Quantile Neural Network Framework for Two-stage Stochastic Optimization: Application to Power Systems Operation.*
Jun. 2023	20th Int. Conf. on Constraint Programming, AI and OR (CPAIOR), Nice, France *A Neural Network-Based Distributional Constraint Learning Methodology for Mixed-Integer Stochastic Optimization.*

FUNDED PROJECTS

2025 – Present	*ODEON – Federated Data and Intelligence Orchestration & Sharing for the Digital Energy TransitioN* Horizon Europe · Role: Researcher
2024 – Present	*Algorithms for Stochastic Optimization Using Data-driven and Learning Analysis* Spanish State Research Agency (PID2023-151013NB-I00) · Role: Researcher
2024 – Present	*SOLAROPIA-CM-UC3M – Operación e Integración de las Plantas Solares de Torre en el Sistema Eléctrico Español* Comunidad de Madrid · Role: Researcher
2021–2024	*Optimization under Uncertainty and Stochastic Control* Spanish State Research Agency (PID2020-116694GB-I00) · Role: Researcher
2021	*Probabilistic Forecasting and Meta-heuristic Optimization for Solar/Wind Resources in the Iberian Peninsula* Spanish State Research Agency (PID2019-107455RB-C22) · Role: Research Assistant

SOFTWARE

CONDUCTOR	github.com/antonioalcantaramata/CONDUCTOR · Main developer, 2026 LLM-orchestrated digital twin for uncertainty-aware power-system operations, pairing a pandapower/Pyomo analysis backend with a natural-language agent.
Quantile2SP	github.com/antonioalcantaramata/Quantile2SP · 2025 Quantile neural network surrogates for two-stage stochastic programs.
CCL-Tool	github.com/antonioalcantaramata/ccl_tool · 2023 Chance-constraint learning and its integration with mixed-integer optimization.
POPI-HN	github.com/antonioalcantaramata/POPI-HN · 2022 PyTorch framework for hypernetwork-based Pareto optimal prediction intervals.

LANGUAGES & COMPUTER SKILLS

Languages	Spanish (native), English (fluent), Italian (basic)
Programming	Python, R, MATLAB
Libraries	PyTorch, scikit-learn, Gurobi, Pyomo, pandapower